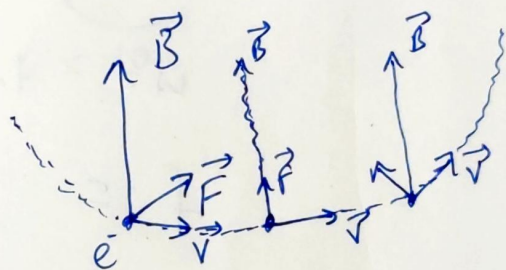
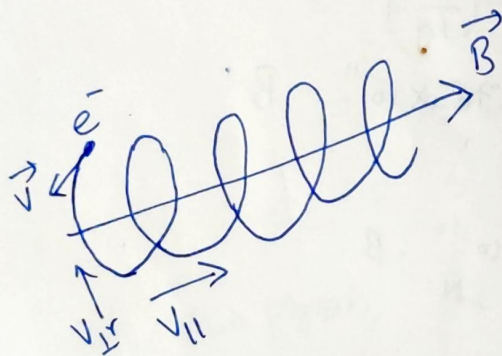


30/11/26

Cyclotron Radiation

Particle — non relativistic electron
moving in a magnetic field

No electric field's acting on the
particle because all this is
happening at near vacuum.



↑
particle goes around
in circle

∴ centripetal force = magnetic force.

Lorentz's force equation

$$m \frac{v_\perp^2}{r} = -q (\vec{v}_\perp \times \vec{B})$$

$$\text{as } \vec{v} \times \vec{B} \rightarrow \vec{v}_\perp \times \vec{B}$$

$$\Rightarrow \frac{m v_\perp^2}{r} = q (\vec{v}_\perp \cdot \vec{B})$$

* $v_\perp$ is ~~along~~ $\perp$ to $\vec{B}$ → this determines the circular motion

* $v_\parallel$ shows how the e^- drifts along the helix

But as the electron is going around in a circle, the angular velocity can be written as velocity v / radius $\Rightarrow \omega = v/r$

$$\omega = \frac{v}{r}$$

$$\Rightarrow m \frac{v_{\perp}}{r} = e v_{\perp} B$$

$$m \cdot \omega = eB$$

$$\Rightarrow \omega = \frac{eB}{m_e}$$

$\left(\frac{e}{m_e}\right) \rightarrow$ electron charge to mass ratio

$$\omega = 1.76 \times 10^{11} \left(\frac{C}{kg}\right) \cdot B (T)$$

$$f = \frac{1}{2\pi} \cdot 1.76 \times 10^{11} \cdot B$$

$$f = 2.8 \times 10^{10} \cdot B$$

$\Rightarrow$ $\downarrow$
cyclotron freq.

depends only on the magnetic field.

Observation:

- as the ω_c/f_c depend only on the magnetic field, for a constant B , ω_c will also be constant. This implies that the radius has to increase if a particle entered in with a larger velocity.

$$\omega = \frac{v_{\perp}}{r}$$

if v_{particle} was $2v_{\perp}$, and ω has to be $\frac{2v_{\perp}}{r}$
same, then $r = \frac{2v_{\perp}}{\omega} \rightarrow \frac{r_2}{r_1} = \frac{2v_{\perp}}{v_{\perp}} = 2$

• Example ①

- at a height of $1.5 R_{\odot}$ above the surface of the sun [the corona], the magnetic field is 2 G. [$1 \text{ G} = 10^{-4} \text{ T}$]. what is the cyclotron freq?

Ans

$$f_c = 2.8 \times 10^{10} \cdot B$$

$$= 2.8 \times 10^{10} \cdot 2 \times 10^{-4}$$

$$= 5.6 \times 10^6$$

$$f_c = 5.6 \text{ MHz}$$

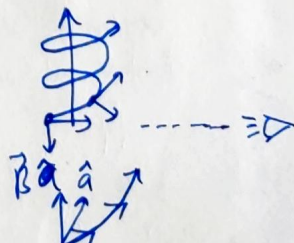
- at sun spots, the field can reach upto 1000 G. What is the cyclotron frequency?

Power from cyclotron emission

Start with Larmor formula:

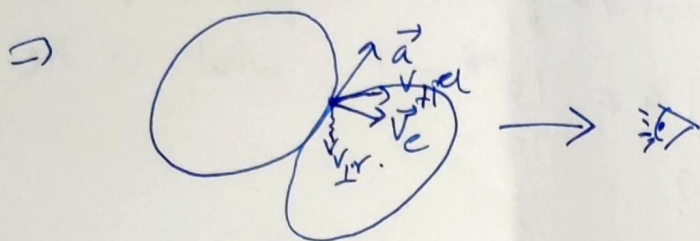
$$P = \frac{q^2 a^2}{6\pi\epsilon_0 c^3}$$

This is valid as an electron is moving in a straight line. acceleration vector is towards the $\vec{F}$.



This would look like a ~~the~~ particle moving on ~~the~~ a ~~loop~~ Helixian dipole.

④ So, the power would be harnessed like for an electron ~~non~~ accelerated.



In cyclotron the acceleration just changes the direction of the electron due to the $\vec{B}$ field and not its velocity itself.

But $F = m_e a$

⇒ $e \cdot v_{\perp} B = m_e a$

$\frac{e v_{\perp} B}{m_e} = a$

⇒ we can plug this into the power equation

$$P = \frac{e^2 a^2}{6\pi\epsilon_0 c^3} = \frac{e^2}{6\pi\epsilon_0 c^3} \left[\frac{e v_{\perp} B}{m_e} \right]^2$$

$$= \frac{e^4 v_{\perp}^2 \cdot B^2}{6\pi\epsilon_0 c^3 m_e^2}$$

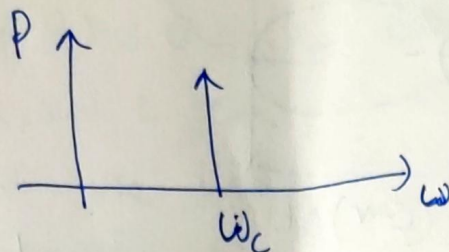
However, $\frac{eB}{m_e} = \omega = \omega_c$

⇒ $P = \frac{e^2}{6\pi\epsilon_0 c^3} \left[\frac{eB}{m_e} \right]^2 \cdot v_{\perp}^2$

$P = \frac{e^2}{6\pi\epsilon_0 c^3} \cdot \omega_c^2 \cdot v_{\perp}^2 \Rightarrow P \propto \omega_c^2 \propto B^2$

cyclotron freq

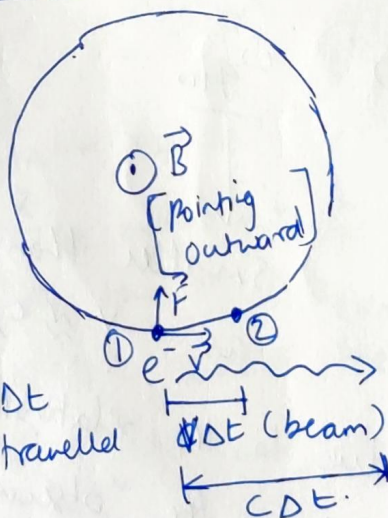
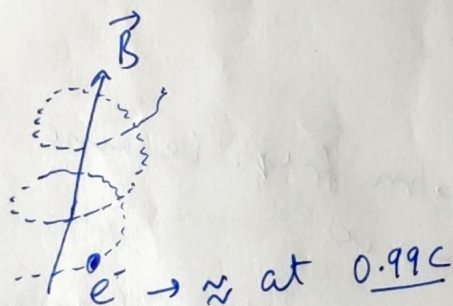
Cyclotron power spectrum



Synchrotron Radiation

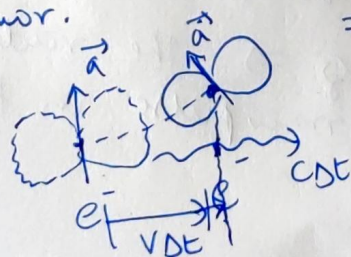
* In cyclotron the electrons are mildly relativistic
 $[0.2c \text{ to } 0.9c]$.

* In synchrotron, the electrons are relativistic
 $\Rightarrow$ velocities are $\sim \underline{0.99c}$.



electron has now moved $v\Delta t$
 & the radiation would've travelled $c\Delta t$.

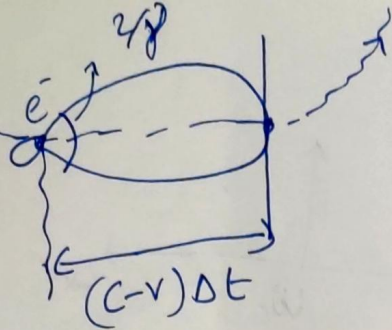
But the electron at pos 1 would have emitted like in Larmor.



$\Rightarrow$ The back lobe of the radiation is now at the front.

$$\underline{(c-v)\Delta t}$$

due to this,
the radiation
appears beamed
when seen from
the observer.



But when electrons are travelling at relativistic speeds, then the mass of the electron will change by the Lorentz factor, γ

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

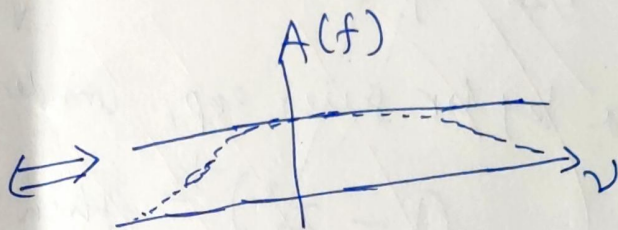
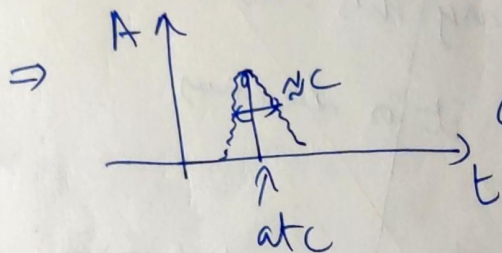
$$\Rightarrow \omega_{\text{gyro}} = \frac{eB}{\gamma m_e}$$

Smaller than cyclotron freq. because
as $v \approx c$, $\gamma \rightarrow \infty$

But, due to the relativistic motion, the ~~same~~ frequency observed by the "observer" will be different from ω_{gyro} .

- ① Radiation is beamed with a cone angle $2/\gamma \Rightarrow$ the radiation will be a narrow pulse in time

as the electron is travelling close to speed of light, $(C - v) \approx 0 \Rightarrow$ the ~~radiation~~ distance b/w the forward & end of the pulse is shortened $\Rightarrow$ the width of the pulse becomes narrower.



Quantitatively, ~~the~~ for one cycle $2\pi/\gamma$, the radiation will reach the observer at

$$\Delta t_{\text{beam}} = \frac{1}{2\pi} \left(\frac{2\pi}{\gamma} \right) \left[\frac{2\pi}{\omega_{\text{gyr}}} \right] = \frac{2}{\gamma \omega_{\text{gyr}}}$$

$\downarrow$
gyration period.

But the front of the beam ~~to the~~ moves at C , & the electrons move at v . So, by the time of Δt_{beam} , the radiation has travelled $C \Delta t$, while the electron has travelled $v \Delta t$. We can get the pulse width from this as:

$$\Delta t_{\text{pulse}} = \frac{C \Delta t_{\text{beam}} - v \Delta t_{\text{beam}}}{C}$$

$$\Delta t_{\text{pulse}} = \left(1 - \frac{v}{C} \right) \Delta t_{\text{beam}}$$

Now we know $\gamma =$

$$\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Squaring on both sides,

$$\gamma^2 = \frac{1}{1 - \frac{v^2}{c^2}}$$

Simplifying, ~~2~~

$$1 - \frac{v^2}{c^2} = \frac{1}{\gamma^2}$$

$$\Rightarrow \frac{v^2}{c^2} = 1 - \frac{1}{\gamma^2}$$

Sq. rt. on both sides will give

$$\frac{v}{c} = \sqrt{1 - \frac{1}{\gamma^2}}$$

this can be Taylor series expanded to get

$$\frac{v}{c} \approx 1 - \frac{1}{2\gamma^2}$$

$$\Rightarrow 1 - \frac{v}{c} = \frac{1}{2\gamma^2}$$

$$\Rightarrow \Delta t_{\text{pulse}} = \frac{1}{2\gamma^2} \cdot \Delta t_{\text{beam}}$$

$$\Delta t_{\text{pulse}} = \frac{1}{2\gamma^2} \cdot \frac{2\pi}{\gamma \omega_{\text{gyr}}} \approx \frac{1}{\gamma^3 \omega_{\text{gyr}}}$$

$$\therefore \Delta t_{\text{pulse}} = \frac{1}{\gamma^2} \cdot \frac{m_e c}{eB}$$

Synchrotron frequency is then

$$\nu_{\text{synch}} \approx \frac{1}{\Delta t_{\text{pulse}}} \approx \frac{\gamma^2 eB}{m_e c}$$

But the relativistic energy is

$$E = \gamma m_e c^2 \Rightarrow \gamma^2 = \frac{E^2}{m_e^2 c^4}$$

$$\Rightarrow \nu_{\text{synch}} \approx \frac{E^2}{m_e^2 c^4} \cdot \frac{eB}{m_e c}$$

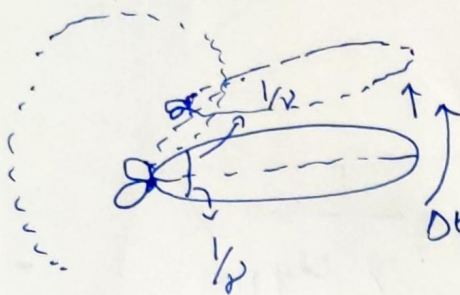
$$\left\{ \nu_{\text{synch}} \approx \frac{E^2}{m_e^2 c^4} \cdot \frac{eB}{m_e c} \approx \frac{E^2 eB}{m_e^3 c^5} \right\}$$

↓
Synchrotron peak

2/2/20

(1)

Synchrotron: frequencies & Power



* as the beam goes around in a spiral due to the electron motion, the time it takes for the beam to sweep a observer at '∞' is: Δt_{beam}

we write $\omega_{gyro} = \frac{eB}{\gamma m_e}$

∴ total time it takes to go a full spiral is:

$$T_{gyro} = \frac{2\pi}{\omega_{gyro}}$$

Out of this, the beam sweeps across an observer

for $\Delta t_{beam} = \frac{\text{beam angle}}{2\pi} \cdot T_{gyro}$

$$= \frac{1}{2\pi} \cdot \frac{2}{\gamma} \cdot \frac{2\pi}{\omega_{gyro}} = \frac{2}{\gamma \omega_{gyro}}$$

$$\Delta t_{beam} = \frac{2}{\gamma \omega_{gyro}}$$

but the pulse will be seen by the observer for $(c-v)\Delta t_{beam}$ duration. because the electron catches up with the radiation

$$\therefore \Delta t_{\text{pulse}} \approx \left(1 - \frac{v}{c}\right) \cdot \Delta t_{\text{beam}}$$

$$\Delta t_{\text{pulse}} \approx \frac{1}{2\gamma^2} \cdot \Delta t_{\text{beam}}$$

$$\approx \frac{1}{2\gamma^2} \cdot \frac{2}{\gamma \omega_{\text{gyr}}} \approx \frac{1}{\gamma^3 \omega_{\text{gyr}}}$$

$$\Rightarrow \Delta t_{\text{pulse}} \approx \frac{1}{\gamma^2} \cdot \frac{m_e c}{e B}$$

$$\text{As } v \approx c, \gamma \uparrow \Rightarrow \Delta t_{\text{pulse}} \rightarrow 0$$

Synchrotron frequency

$$\nu_{\text{Synch}} \approx \left\{ \frac{1}{\Delta t_{\text{pulse}}} \right\} \approx \frac{\gamma^2 e B}{m_e c}$$

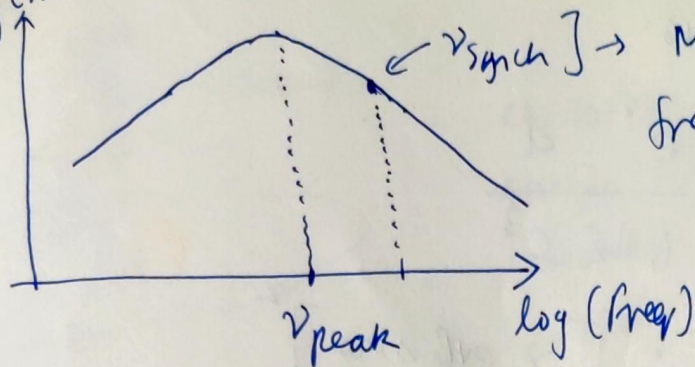
$$\text{But } \gamma = \frac{E}{m_e c^2} \Rightarrow \gamma^2 = \frac{E^2}{m_e^2 c^4}$$

$$\therefore \nu_{\text{Synch}} \approx \frac{E^2}{m_e^2 c^4} \cdot \frac{e B}{m_e c} \approx \frac{e E^2 B}{m_e^3 c^5}$$

$$\Rightarrow \boxed{\nu_{\text{Synch}} \propto E^2 B} \quad \text{as } \frac{e}{m_e^3 c^5} \text{ is constant}$$

↓
Synchrotron characteristic freq

log (mW/m²)



Median of the frequencies where power is contained

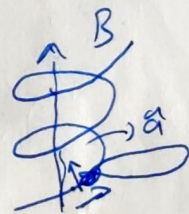
$$\nu_{\text{peak}} \approx 0.0692 \cdot \frac{e}{m_e c} \cdot \gamma^2 B$$

$$\approx 0.0692 \frac{e}{m_e^3 c^5} E^2 B$$

$$\nu_{\text{peak}} \approx 0.0692 \nu_{\text{sync}}$$

Total power for Synchrotron emission by single electron

$$a_{\text{acc frame}} = \gamma^2 a_{\text{obs. frame}}$$



beaming $\rightarrow$ radiation is detected in electron's direction of motion, and is tangential to electron's gyration motion

$$\gamma m_e \vec{a}_{\text{obs}} = e \vec{v} \times \vec{B}$$

$$\gamma m_e \vec{a}_{\text{obs}} = e v B \sin \theta$$

$$\vec{a}_{\text{obs}} = \frac{e v B \sin \theta}{\gamma m_e}$$

$$a_{\text{elect. frame}} = \gamma^2 \frac{e v B \sin \theta}{\gamma m_e} = \left[\frac{\gamma \cdot e v B \sin \theta}{m_e} \right]$$

But Larmor eqn: says.

$$P = \frac{e^2 \cdot a^2}{6\pi\epsilon_0 c^3}$$

$$= \frac{e^2}{6\pi\epsilon_0 c^3} \left[\frac{\gamma v B \sin\theta}{m_e} \right]^2$$

$$= \frac{e^2}{6\pi\epsilon_0 c^3} v \cdot \gamma^2 \cdot \frac{e^2 B^2 \sin^2\theta}{m_e^2}$$

$$= \frac{e^4}{6\pi\epsilon_0 c^3} \cdot \frac{E^2}{m_e^2 \cdot c^4} \cdot B^2 \sin^2\theta$$

$$\boxed{P \propto E^2 \cdot B^2 \sin^2\theta}$$



① Larger power is radiated for electrons with higher energy

② Larger power radiated ⁱⁿ for regions of high $\vec{B}$ field.

③ particle $\parallel$ to $\vec{B}$ field emits nothing, particle $\perp$ to $\vec{B}$ field emits everything, but when it is b/w $0 - 90^\circ$, it does ~~the~~ a helical motion.



As the particle moves to a stronger $\vec{B}$ field, the magnetic moment μ has to be constant.

$$\Rightarrow \mu = \frac{\frac{1}{2} m v_{\perp}^2}{B} = \text{const.}$$

Since $v_{\perp} = v \sin \alpha$

$$\frac{\sin^2 \alpha}{B} = \text{constant.}$$

$\Rightarrow$ as B increases, α must increase also.

$\Rightarrow$ as $\alpha \uparrow$, spiral gets flattened out.

Hence, gyro radius = $\frac{m v_{\perp}}{e B}$

$$\Rightarrow \text{as } B \uparrow, v_{\perp} = \sqrt{\frac{2 \mu B}{m}}$$

$$\Rightarrow r_g = \frac{m}{e B} \sqrt{\frac{2 \mu B}{m}} \Rightarrow r_g \propto \frac{1}{\sqrt{B}}$$

$\omega_g \propto B \Rightarrow \omega_g \uparrow$ close to the regions
~~mirror region~~ where $B \uparrow$.

But radiation happens due to a population of electrons at different energies. When there is a broad range of energies, the emitted frequencies will also be broad.

Total number of electrons in the energy range

$$E \text{ to } E + dE \rightarrow N(E)dE$$

$\Rightarrow$ Power emitted by electrons in this energy range will be $\rightarrow$ Energy distribution of radiating electrons.

$$P(\nu) d\nu \propto \boxed{N(E) dE} E^2 B^2$$

$E \rightarrow$ Energy of electron whose peak is ν .

But at higher energies fewer e^- may actually get accelerated.

$$\Rightarrow N(E) dE = N_0 \cdot E^{-p} \cdot dE$$

$$\therefore P(\nu) d\nu \propto N_0 \cdot E^{-p} \cdot dE \cdot E^2 \cdot B^2$$

$$P(\nu) d\nu \propto N_0 \cdot E^{-(p-2)} dE \cdot B^2$$

But to ensure that the integrals add up the
lets us see

Let us define the power as a power law curve
changing with freq

$$\Rightarrow P(\nu) \propto \nu^\alpha$$

$$\Rightarrow \nu^\alpha \cdot d\nu \propto E^{(2-\alpha)} \cdot dE$$

but from Sum ch. characteristic freq eqn,

$$\nu \propto E^2$$

$$\Rightarrow d\nu \propto 2E dE -$$

$\therefore$ Sub this in the prev. eqn

$$E^{2\alpha} \cdot 2E dE \propto E^{2-\alpha} \cdot dE$$

$$2E^{(2\alpha+1)} dE \propto E^{2-\alpha} dE$$

equating the powers,

$$2\alpha+1 = 2-\alpha$$

$$2\alpha = 2-\alpha-1$$

$$2\alpha = -\alpha+1$$

$$\Rightarrow \left\{ \alpha = \frac{1-\alpha}{2} \right\}$$

↓

relation b/w frequency variation of
intensity with electron energy
dist. power law index

α of radio sources

↓

Radio loud galaxies.

vary b/w $\alpha \sim -0.4$ to -1.0

-0.7 to -0.8 are ~~common~~ most common

$\Rightarrow p \sim 1.8$ to $3 \rightarrow$ energy distribution becomes steeper.

Application to Cygnus-A

$$S_\nu \propto \nu^{-\alpha}$$

$$\alpha = 0.7 - 0.8$$

$$p = 2.4 - 2.6$$

$$\Rightarrow N(E) dE \propto E^{-2.5} dE$$

has more low energy relativistic electrons than high energy.

Core spectra is flat? Explain using Inverse

Synchrotron $\rightarrow$ energy loss

Inverse Compton scattering $\rightarrow$ energy loss.

$$\Rightarrow \frac{dE}{dt} = -a E^2 ; \alpha \propto (B^2 + B_{IC}^2)$$

$$B_{IC} \approx 3.6 \text{ MG.}$$

5/2/26: ~~Radio~~ Application of Synchrotron in understanding Cygnus-A

Flux of Cygnus $\propto \nu^{-0.85}$ Integrated over the source

But, ~~if~~

But if we make high resolution images of the source, we see that there is spatial and spectral variations in the Cygnus-A spectrum.

Spectral variations

at low freq,

$$S_\nu \propto \nu^{-\boxed{0.71}} \rightarrow \text{steep spectrum}$$

at high freq

$$S_\nu \propto \nu^{-\boxed{1.06}}$$

- So, high freq is fitted well by a steeper power law $\Rightarrow$ intensities fall more at high freq
- Falls slower at lower freq, This "turnover" in freq occurs close to $\sim 1 \text{ GHz}$

if $\alpha \rightarrow$ Intensity Spectral Index it is related to electron energy Spectral Index as:

$$\alpha = \frac{1-P}{2}$$

$$\Rightarrow \cancel{2\alpha} P = -(2\alpha - 1) = 1 - 2\alpha$$

$$\therefore P_{-1.06} = 1 - 2(-1.06) \sim \boxed{3.12} \rightarrow \text{steep! old electrons}$$

$$P_{-0.9} = 1 - 2(-0.95) \sim 2.7$$

$$P_{-0.71} = 1 - 2(-0.71) \sim 2.6 \rightarrow \text{relatively young.}$$

$\rightarrow$ What causes this Spectral variation?

$\rightarrow$ Where do the electrons come from?

Refer to Cygnus A's image.

To understand the changes in the Spectral Indices itself, we will look at the energy loss rates:

$$P = -\frac{dE}{dt} = \frac{2e^4 B^2}{3mc^2} \gamma^2 \beta^2$$

$$-\frac{dE}{dt} \propto \gamma^2 \frac{E^2}{m^2 c^4} \quad \frac{dE}{dt} \propto B^2$$

$$\Rightarrow \frac{dE}{dt} \propto E^2, \quad \frac{dE}{dt} \propto B^2$$

$$P_{\text{Synch}} = \frac{e^4}{6\pi\epsilon_0 c^3} \cdot \frac{E^2}{m_e^2 c^4} \cdot B^2 \sin^2 \alpha \cdot v^2$$

$$P_{\text{Synch}} = \frac{e^4}{6\pi\epsilon_0 m_e^4 c^7} E^2 B^2 \sin^2 \alpha \cdot v^2$$

$$\text{but as } v = c, P_{\text{Synch}} = \frac{e^4}{6\pi\epsilon_0 m_e^4 c^5} E^2 B^2 \sin^2 \alpha$$

but $P = -\frac{dE}{dt}$, from this we can get the

$$t_{\text{syn}} = \frac{\text{Energy}}{\text{Energy radiated/power}} = \frac{E}{\frac{e^4}{6\pi\epsilon_0 m_e^4 c^5} E^2 B^2 \sin^2 \alpha}$$

$$\Rightarrow t_{\text{syn}} = \frac{6\pi\epsilon_0 m_e^4 c^5}{e^4 E B^2 \sin^2 \alpha}$$

We see that the plot (on screen) that the spectral index for Cyg - A changes from -0.8 to -1.06 at $\nu \sim 19 \text{ Hz}$. This frequency is called the break frequency ν_b .

$$\text{But } t_{\text{syn}} \propto B^{-2} E^{-1}$$

But $\nu_{\text{synch}} \propto B E^2$] $\rightarrow$ we can apply this to the break freq., as the break freq is the

- characteristic freq. of a new electron population now.

$$\Rightarrow \nu_b \propto E^2 B$$

$$\therefore E \propto \sqrt{\frac{\nu_b}{B}}$$

Plug this in exp for T_{synch}

$$T_{\text{synch}} \propto B^{-2} \cdot E^{-1}$$

$$\propto B^{-2} \left(\left(\frac{\nu_b}{B} \right)^{1/2} \right)^{-1} \Rightarrow B^{-2} \cdot B^{1/2} \cdot \nu_b^{-1/2} \\ \propto B^{-3/2} \cdot \nu_b^{-1/2}$$

Including the constant, we can write T_{synch}

$$\text{as } T_{\text{synch}} \approx 5.06 \times 10^5 \frac{B^{1/2}}{B^2 + \underbrace{B_{\text{CMB}}^2}} \cdot \nu_b^{-1/2}$$

because the electrons
lose energy by scattering
off CMB photons.

Inverse Compton Cooling

How do we get B ? from assumption of
equipartition of energy, ~~it can simply be~~
written as,

$$B_{\text{min}} = 4.5 \times 10^{-4} \text{ gauss}$$

B for Cyg-A & Luminosity & its radial extent as

$$B \propto \left(\frac{L}{2}\right)^{2/7} \cdot R^{-6/7}, \text{ assuming spherical symmetry.}$$

$$B \approx 10^{-4} \text{ gauss.}$$

Similarly minimum energy for a source like Cygnus for given luminosity is

$$E \propto L^{4/7} \cdot R^{9/7} \sim 5 \times 10^{53} \text{ J}$$

$$L \approx 4\pi d^2 S_\nu \left(\frac{\nu}{\text{Hz}}\right)$$

$$\approx 4\pi d^2 \cdot 2000 \text{ Jy} (\nu)^{-0.8}, \quad \nu \text{ in GHz}$$

$$\therefore L_{\nu} \approx \left(7.71 \times 10^{24}\right)^2 4\pi \cdot \nu^{-0.8}$$

$$L_{\nu} = 1.49 \times 10^{28} \cdot \nu^{-0.8} \text{ W}$$

$$L = \int_{10 \text{ MHz}}^{100 \text{ GHz}} L_{\nu} \cdot d\nu \approx 1.49 \times 10^{28} \int_{10^7}^{10^{11}} \nu^{-0.8} d\nu$$

$$\approx 1.49 \times 10^{28} \left[\frac{\nu^{-0.8+1}}{-0.8+1} \right]_{10^7}^{10^{11}}$$

$$\approx 1.49 \times 10^{28} \cdot \frac{\nu^{0.2}}{0.2}$$

$$\approx 1.49 \times 10^{28} \left[\frac{(10^{11})^{0.2} - (10^7)^{0.2}}{0.2} \right]$$

$$L \approx 9.9 \times 10^{30} \text{ W} \sim 1.46 \times 10^{38} \text{ W}$$

This is ~ 11 times higher than the bolometric luminosity of the Sun!

Now, B_{cmb} can be neglected for Cyg A as

$$B_{\text{cmb}} \approx 0.36 \text{ nT at Cyg A redshift}$$

$$\approx 3.6 \text{ } \mu\text{G} \quad \underline{\text{Cyg A } (z \sim 0.06)}$$

$\Rightarrow$ plugging this into τ_{synch} ,

$$\tau_{\text{synch}} \approx \frac{E_{\text{min}}/(1+\eta)}{L} \approx \frac{5.4 \times 10^{52} / (1+\eta)}{1.46 \times 10^{38}}$$

$$\approx \frac{5.4 \times 10^{52} (1+\eta)^{4/7}}{1.46 \times 10^{38} (1+\eta)}$$

$$\approx 3.6 \times 10^{14} (1+\eta)^{-3/7} \text{ s} \approx 3.6 \times 10^{14} \text{ s}$$

$\eta \rightarrow$ varies from $\frac{1}{2}$ to $\boxed{2000} \rightarrow$ accelerated protons, hot emitting radiation
for e^- , positron plasma

$$\Rightarrow 3.6 \times 10^{14} (1+\eta)^{-3/7} \approx 2.6 \times 10^{14} \text{ s}$$

$$\quad \quad \quad \underbrace{(2 \text{ or } 2001)^{-3/7}}_{\text{or}} \approx 1.3 \times 10^{13} \text{ s}$$

$$\boxed{\tau_{\text{synch}} \sim 1 \text{ Myr}}$$

the electrons (emitting synchrotron radiation) travel along the jet, as can be seen from the steep spectral index along the jet, and reach the ends of the lobes, ~~at 0.25 ly~~ ~ 0.25 ly.

~~$\Rightarrow$ Plasma/material must be accelerated to populate~~

$$\therefore \text{Speed of Jet} \sim \frac{0.25 \text{ ly}}{1 \text{ Myr}} \sim \underline{0.205c}$$

The ~~gas~~ e^- gets reaccelerated at the boundary which causes them to gain energy & create hot spots indicated by the spectral index there.

Mechanism of acc etc, for another time.

Synchrotron Self Absorption

We see that as there will be more & more e^- available at low freq, ~~the~~ we may be tempted to think that sources may be coldest bright at low freq.

But the emitting particles sometimes in in local Thermodynamic Equilibrium. $\Rightarrow$ They are like a thermal source & their T_b cannot be greater than their kinetic temperature (as they are thermal)

This causes Synchrotron self-absorption, leading to ~~to absorption of the emitted~~, preventing the brightness from increasing.

Most e^- with energy $E = \gamma mc^2$ emit at critical freq at which electron at temp T , cannot emit more than T

$$\gamma_c \sim \frac{\gamma^2 e B}{2\pi mc}$$

$$\gamma \approx \left(\frac{2\pi mc \nu_c}{eB} \right)^{1/2} \quad \text{[Second law]}$$

Electrons at temp T' cannot supply more heat than T'

But effective temp of relativistic e^- with energy E is

$$T_e = \frac{E}{3k} = \frac{\gamma mc^2}{3k} = \left(\frac{2\pi mc \nu_c}{eB} \right)^{1/2} \frac{mc^2}{3k}$$

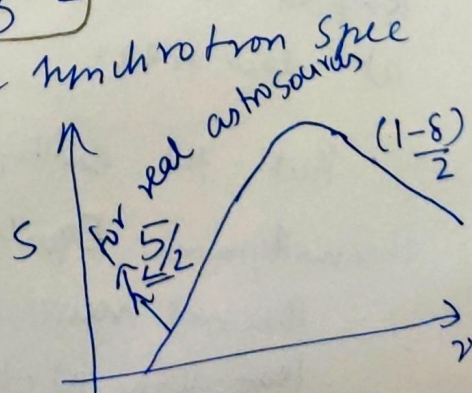
But at ν sufficiently low freq, the T_e will approach T_e . Sub T_e in Rayleigh Jeans' law,

$$I_\nu = \frac{2KT}{\lambda^2} \approx \frac{2KT_e \nu^2}{c^2}$$

$$\approx \frac{2k\nu^2}{c^2} \cdot \left(\frac{2\pi mc \nu_c}{eB} \right)^{1/2}; \quad \text{if } \nu \sim \nu_c$$

$$I_\nu \propto \gamma^{5/2} \cdot B^{-1/2}$$

$\therefore$ at low freq, self absorbed synchrotron spec has a $\nu^{5/2}$ dependence $\rightarrow$ steep slope at the turnover freq, $\tau = 1$



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Nature of radio signals

A "lot" of electrons contribute to all radio emission

⇒ Makes them only partially coherent at best

* Exceptions: MASER, Plasma etc

∴ all electrons emit at different energies.
[It's a spectra in synchrotron, or in thermal noise]

If Thermal

↓
Rayleigh Jeans Law = $\frac{2kT_{obj}}{\lambda^2}$

If non thermal

$$S_\nu \propto \nu^{-\alpha}$$

but Specific Intensity, $S_\nu \propto \langle \vec{E} \cdot \vec{E}^* \rangle$

$P_{flux} = \int S_\nu \cdot d\Omega$ (to some extent)
↳ encountered by an antenna with a collecting area A.

$$S_\nu \rightarrow \text{W m}^{-2} \text{ Hz}^{-1} \text{ Sr}^{-1}$$

↓
antennas have a field of view

From ~~power~~ flux, S_ν , we can go to power measured by antenna as

$$- S_\nu \cdot A_e$$

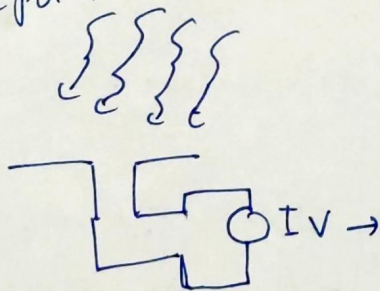
effective area of the antenna

$$\Rightarrow \text{PSD} = S_\nu A_e$$

$$\text{PSD} = \int_r I_\nu \cdot d\Omega \cdot A_e$$

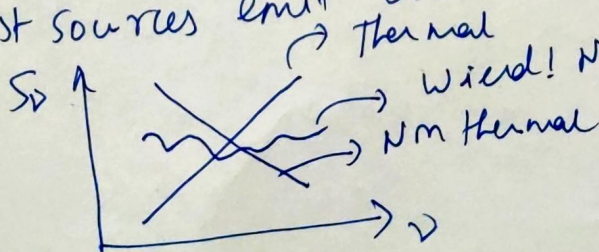
$[W/Hz]$

But in radio astronomy, antennas can measure a power that is proportional to its voltage at its feed point.



~~But~~ However, the ~~power~~ fields measured at the antenna are superposition of fields from the source at a specific frequency.

And most sources emit over wide-band.



If we are trying to measure Power Spectral density, PSD must be of similar ~~sig~~ shape.

$$\Rightarrow \text{PSD} \propto I_\nu$$

But we get to PSD from voltages after:

~~$\text{PSD} = \int_{-\infty}^{\infty} \langle E(t) E(t+\tau) \rangle d\tau$~~

$$\text{PSD} = \underbrace{\langle E(\nu) \cdot E^*(\nu) \rangle}_{\text{averaged over time.}} \rightarrow \text{second order moment.}$$

~~But~~ If we think about the nature of $E(\nu)$ itself,
 $E(\nu) \rightarrow$ is ~~is~~ a superposition of contributions from different fields, each generated by a different electron!

$\Rightarrow$ Each field has a different field Amplitude

$\Rightarrow$ Each has a different phase, both temporal & spatial!

In a sufficiently large collection of electrons,
 A is a random function with '0' mean
 ϕ is a random function with '0' mean

$\Rightarrow$ we are not measuring the field themselves, but rather their:

- Mean values
- Fluctuations



How do we quantify these?

Answer: Statistics.

What do we know about the fields?

① They are zero mean.

② If they are not perturbed, their mean will not change $\Rightarrow$ they are stationary in wide sense.

Every time when a source is observed for a ~~duration~~ long duration, then we measure a part of its "ensemble". Between one part of the "ensemble" and the other part, when an ^{Expectation} ~~averaging~~ is done, the ~~avg~~ expectation values are equal.

These properties also make the signal "Ergodic"
For an ergodic process, the sample average = the time average.

This property of the signal is why radio astronomy works!

So in effect, we are studying only the statistics of the signal.

To quantify these statistics we can start writing about the signal using what we know about it:

- ① Its average level,] $\rightarrow$ from the histogram
- ② Its ~~the~~ fluctuations]

The average or the mean level is $\rightarrow \mu$

The variation of the signal can be quantified as $\rightarrow \sigma$

But to design a receiver, we need to get some info about the signals themselves.

$\Rightarrow$ we need to translate these random numbers into something non random

We will use expectation values for these.

But as these values are random, and their histogram is gaussian, we can also write ~~a~~ the "probability" with which we will measure the value for a certain sample using the probability density function.

$$P(a \leq x \leq b) = \int_a^b P_x(x) \cdot dx$$

Probability that we will measure values between a and b .

when $a = -\infty$, $b = +\infty$, PDF $\rightarrow 1$

If you look in the histogram we see that the ~~highest probability density is close to~~ there is a maximum. This maximum occurs around something called the Expectation value.

When I have done close to ∞ draws, the weighted average of all my draws forms the "Expectation".

$$E[x] = \int x P_x(x) dx$$

Another property of the curve we see is its width. This width shows how ~~my~~ my distribution is spread out.

as it is more spread out, i can ~~draw~~ "probably" draw values away from the mean. The quantity that defines this is called the variance.

Variance is defined as

$$\sigma^2 = \text{Var}(X) = \int (x - \mu)^2 P_X(x) dx$$

$$\mu = 0$$

$$\text{Var}(X) = \int x^2 P_X(x) dx$$

But note, if I had a series of x^2 values, then the above equation will be the expectation of the x^2 sequence.

If x was a stream of voltages, then $x^2 \propto \text{Power}$.

Now, we know power, P

$$P = V_{\text{rms}} \cdot I_{\text{rms}}$$

as in all ~~cases~~ ^{cases}, we try to measure the voltage across an impedance Z , then

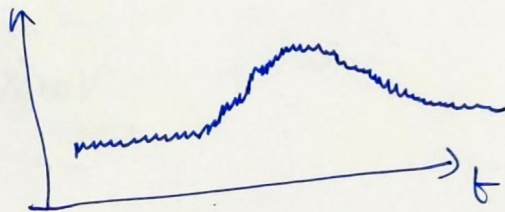
$$P = V_{\text{rms}} \cdot \frac{V_{\text{rms}}}{Z} = \frac{V_{\text{rms}}^2}{Z} = \frac{\sigma^2 / E(x^2)}{Z}$$

$$\Rightarrow \boxed{V_{\text{rms}} = \sqrt{E(x^2)}} \rightarrow \text{True for zero mean processes.}$$

$\therefore$ when $E(x^2) \rightarrow$ high, noise / fluctuations
in the signal is high

This is what happens when a bright source comes in
to the FoV of our telescope

$E(x^2) \uparrow \rightarrow$ Power

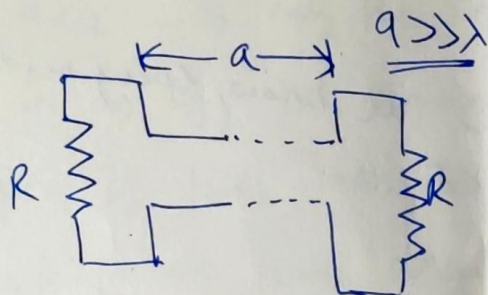


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Thermal noise, its Spectra & Radiometer Equation

[Essential Radio Astronomy, section 2.5]

Motion of charged particles at any temperature above T , generates electrical noise. Spectra is just dependent on T .

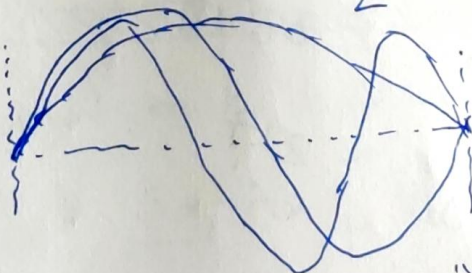


Resistors are matched

if at a wavelength of λ , the standing wave generated in the Transmission line will be at points along the wire with ~~nodes~~

$$a = \frac{n\lambda}{2}, \text{ where } n = 1, 2, 3, 4, \dots$$

because



but, velocity of wave in the "Transmission line" will be decided by the velocity factor and is usually less than c . If v_e is the velocity in the transmission line, then

$$a = \frac{n}{2} \cdot \frac{c}{v} \quad \xrightarrow{\text{replace by } v_e} \quad = \frac{n \cdot v_e}{2v}$$

$$\frac{2va}{v_e} = n$$

∴ The "modes" per frequency are then written

↓
a pattern or a oscillation
that a system settles
into naturally

$$N_D = \frac{2a}{v_e}$$

But from Boltzmann's law we know, every mode has an ^{average} energy $\langle E \rangle = KT$.

$$\Rightarrow E_D = N_D \cdot KT$$

$$E_D = \frac{2aKT}{v_e}$$

As it would take $\Delta t = \frac{a}{v_e}$ for power to flow along this transmission line,

$$E_D = 2 \Delta t KT$$

$$\Rightarrow \frac{E_D}{\Delta t} = 2KT$$

$$P_D = 2KT.$$

∴ power generated by each resistor would then be $P_D = KT$ → like Nyquist's theorem equivalent.

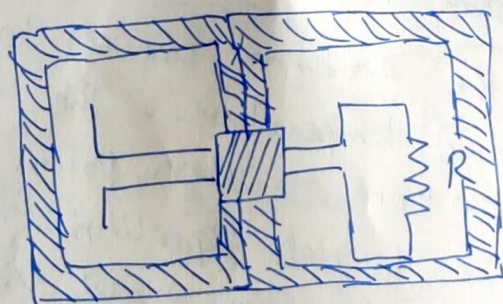
at freq when $h\nu \sim KT$,

$$P_D = KT \left[\frac{h\nu/KT}{\exp(h\nu/KT) - 1} \right]$$

8. A radio telescope in essence can be viewed as using this matched resistor analogy. We will assume that this strawman telescope is just made of a single "dipole" antenna

→ More on this later.

If there was only the sky at a constant temperature, this is as if, the antenna is "dipped" into the sky. If this was in its entirety an isolated system, we can draw it as follows:



The power incident on the antenna per unit Hz is,

$$P_{\nu} = \frac{S_{\nu}}{2} \cdot A_{\nu} \left[\begin{array}{l} S_{\nu}/2 \text{ because for a} \\ \text{single polarization,} \\ \text{we have only half} \\ \text{the power} \end{array} \right]$$

$$\frac{2 P_{\nu}}{S_{\nu}} = A_{\nu}$$

The antenna has an area A_e . This can be direction dependent, may have side lobes, or can just be isotropic in all directions. Either way due to its direction dependence (or independence) we can write an ^{avg.} effective collecting area which is

$$A_e = \frac{\int_{4\pi} A_e(\theta, \phi) \cdot d\Omega}{\int_{4\pi} d\Omega} = \frac{1}{4\pi} \int_{4\pi} A_e d\Omega$$

~~Given that the system~~

Now assume that the antenna is connected to a resistor ^{"load"} which is at the same temperature. The resistor is "matched" to the antenna $\Rightarrow$ Max. Power transfer. As the system is in thermal equilibrium there can be no net power flow, as this would violate the second law of thermodynamics.

$$\therefore \frac{P_r}{T} = \frac{1}{2} \int_{4\pi} A_e \cdot d\Omega \cdot B\nu$$

Power collected
by the antenna
in all directions
due to a
Temp T

However, $B_{\nu} = \frac{2KT}{\lambda^2} \cdot \frac{A_e}{4\pi}$

At low freqs, However, at the power produced by a resistor; and Brightness B_{ν} are, respectively,

$$P_{\nu} = kT \left[\frac{h\nu/kT}{\exp(h\nu/kT) - 1} \right]; B_{\nu} = \frac{2KT}{\lambda^2} \left[\frac{h\nu/kT}{\exp(h\nu/kT) - 1} \right]$$

$\Rightarrow$ If this is sub in prev eqn, then

$$kT = \frac{2KT}{2\lambda^2} \underbrace{\int A_e(\omega, \lambda) d\Omega}_{4\pi \langle A_e \rangle}$$

$$kT = \frac{2KT}{2\lambda^2} \langle A_e \rangle$$

$$\left[\frac{\lambda^2}{4\pi} = \langle A_e \rangle \right] \Rightarrow \text{all lossless antennas have the same average collecting areas}$$

$\langle A_e \rangle \propto \lambda^2 \Rightarrow$ at low freqs, ~~large~~ dipoles made of just thin wires are sufficient to reach very large collecting areas.

For the lossless antenna, we can define a beam solid angle also.

$$\Omega_{\text{beam}} = \int_{4\pi} \frac{A(\theta, \phi)}{A_{\text{max}}} \cdot d\Omega$$

$$\Rightarrow A_{\text{max}} \cdot \Omega_{\text{beam}} = \left\{ \int_{4\pi} A(\theta, \phi) d\Omega \right\} \rightarrow \lambda^2$$

$$A_{\text{max}} \cdot \Omega_{\text{beam}} = \lambda^2$$

Sensitivity equation

Flux measured by a detector, of collecting area A_e like the one defined above is:

$$S_p = \frac{I_p}{4\pi} \cdot \Omega_{\text{beam}}$$

one polarization

at the regime where $h\nu \ll kT$, then

$$I_p = \frac{2kT}{\lambda^2}$$

$$\Rightarrow S_p = \frac{2kT}{\lambda^2} \cdot \Omega_{\text{beam}}$$

$$S_p = \frac{2kT}{A_e}$$

where $T \Rightarrow T_{\text{sys}}$, A_e = collecting area of the receiving element, T_{sys} = System noise temperature.

$$T = \frac{S_p A_e}{2K}$$

Now if $T = T_{\text{sys}}$, then this would be the flux that we will estimate due to the noise temperature of the system itself.

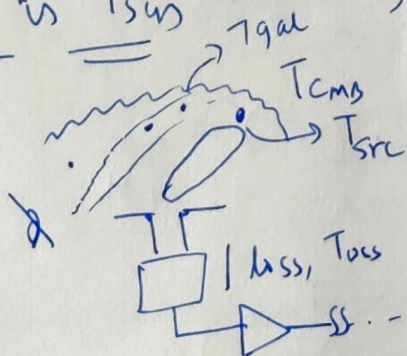
$$\text{SEFD} = \frac{2K T_{\text{sys}}}{A_e}$$

$$T = S_p \left[\frac{A_e}{2K} \right] \rightarrow \text{gain of radio telescope}$$

Change in receiver temperature, per unit flux density.

Maximize $\frac{A_e}{2K}$, minimize SEFD

What is T_{sys} → all other things that the antenna sees other than the source.



$$T_{\text{sys}} = T_{\text{CMB}} + T_{\text{atm}} + T_{\text{gal}} + T_{\text{loss}} + T_{\text{rx}} + \dots$$

When we point to a source, then the noise temperature of the system is:

$$T_{\text{src}} + T_{\text{sys}}$$

Now, we go back to statistics. To recap,

- E-field is a zero mean, σ -variance process

- Each sample is independent of the previous.

- Power $\propto$ Variance of E-field

$\Rightarrow T \propto$ Variance of E-field.

- Any internal change in power $\rightarrow$ change in variance.

- Any source in beam, variance increases.

$\therefore$ Our "signal" is in variance of "E field"

$P_{\text{mean}} \propto \sigma^2 \Rightarrow$ Mean of power is centered in the variance of the fields.

$\therefore$ To measure the mean power accurately, we need to minimize the variance in σ^2 .

But $\text{var}(\sigma^2) \propto \frac{1}{N} \Rightarrow \text{std}(\sigma^2) \propto \frac{1}{\sqrt{N}}$

$$\Rightarrow \frac{\Delta T}{T} \propto \frac{1}{\sqrt{N}} \Rightarrow \Delta T \sim \frac{T_{\text{sys}}}{\sqrt{N}}$$

$\therefore \Delta T$ can be minimized by averaging N samples. If T was T_{sys} , then $\Delta T \sim \frac{T_{\text{sys}}}{\sqrt{N}} \Rightarrow$ Minimum detectable temperature.

If $T = T_{src}$ & T_{sys} = System temperature,

then

$$T_{src} + T_{sys} = T_{meas}^{src} \quad \left[\text{when source is in beam} \right]$$

$$T_{sys} = T_{meas}^{sys} \quad \left[\text{when source is off beam} \right]$$

$$\Delta T_{meas} \propto T_{meas}^{src} - T_{meas}^{sys}$$

$$\Delta T_{meas} \sim T_{src} \rightarrow \text{Signal}$$

$T_{sys} \rightarrow$ RMS fluctuation in system just due to system noise.

$$\sigma(T_{sys}) \rightarrow \text{deviation in } T_{sys} \sim \frac{T_{sys}}{\sqrt{N}}$$

$$SNR \propto \frac{T_{meas}^{src}}{T_{meas}^{sys}} \cdot \frac{1}{\sqrt{N}} \quad \text{If we performed } N \text{ measurements, the}$$

variance in T_{meas}^{sys} $\downarrow$ by $\frac{T_{sys}}{\sqrt{N}}$

$$\Rightarrow \boxed{SNR \propto \frac{T_{meas}^{src} \sqrt{N}}{T_{meas}^{sys}}} \rightarrow \text{Radiometer equation}$$

If measurement performed over a bandwidth Δf , and over a time Δt , then $N \sim \Delta f \cdot \Delta t$.

$$\Rightarrow SNR \propto \frac{T_{src}}{T_{sys}} \sqrt{\Delta f \cdot \Delta t}$$

$$T_{src} \sim \frac{T_{sys} \cdot SNR}{\sqrt{\Delta f \cdot \Delta t}} \rightarrow \text{we can plug this in to SEFD}$$

$$SEFD = \frac{2kT_s}{A_e}$$

$$= \frac{2kT_{\text{sys}}}{A_e} = \frac{2kT_{\text{sys}}}{A_e} \cdot \frac{SNR}{\sqrt{\Delta f \cdot \Delta t}}$$

$\Rightarrow$ Minimum detectable flux at an $SNR = 5$

is $\Rightarrow$ System noise itself

$$\Delta S_{\min} = \frac{2kT_{\text{sys}}}{A_e \sqrt{\Delta f \cdot \Delta t}} \quad (\text{Jy})$$

Question

① Can we keep integrating to ∞ & $\downarrow$
system noise?